Disp & Panel Q&A

Q1: How to troubleshoot the abnormal operation of prog_panel?

The first step is to check whether the hardware wiring is correct. The second step is to check whether the loading screen parameters correspond to the screen in use. You can compare the screen parameters with cat proc/mi_module/mi_panel to check whether the screen parameters are set correctly (refer to "Screen Parameters Configuration" in sigdoc), and check whether the loaded sensor driver is correct. (Refer to the sensor configuration in "Running Environment Configuration"). The third step is to check whether the gstDisplayOpt, gstSensorAttr, SENSOR_IDX_FOR_PANEL, SENSOR_IDX_FOR_HDMI configuration in the demo code is correct, and check whether the initialization of several panel control pins defined by the GPIO_PANEL_ macro is correct.

Q2: In the demo of prog_panel, the default is imx415 to hdmi, imx307 to point screen. Now I want to use imx307 to output hdmi, how do I need to modify it?

The simplest modification is to directly exchange the 0 and 1 defined by SENSOR_IDX_FOR_PANEL and SENSOR_IDX_FOR_HDMI.

Q3: The RST & INT hardware of the touch screen cannot be controlled normally and the touch screen cannot be used normally?

The interrupt and reset (GPIO13, GPIO14) pins of the touch screen of the development board conflict with the JTAG pins (GPIO13~GPIO16), so IPL and IPL_CUST need to use the NO_JTAG version

Q4: Are there abnormal bright spots at the junction of light and dark on the TTL screen?

If the hardware wiring is long, just remove the external extension cable. Or to enhance the driving capability of the TTL output port, refer to the following script:

```
/customer/riu_w 103e 13 D5
/customer/riu_w 103e 14 D5
/customer/riu_w 103e 15 D5
/customer/riu_w 103e 16 D5
/customer/riu_w 103e 17 D5
/customer/riu_w 103e 18 D5
/customer/riu_w 103e 19 D5
/customer/riu_w 103e 1A D5
/customer/riu_w 103e 43 D5
/customer/riu_w 103e 44 D5
/customer/riu_w 103e 45 D5
/customer/riu_w 103e 46 D5
/customer/riu_w 103e 47 D5
/customer/riu_w 103e 48 D5
/customer/riu_w 103e 49 D5
/customer/riu_w 103e 4A D5
/customer/riu_w 103e 4B D5
/customer/riu_w 103e 4C D5
/customer/riu_w 103e 5F D5
/customer/riu_w 103e 60 D5
/customer/riu_w 103e 61 D5
/customer/riu_w 103e 62 D5
/customer/riu_w 103e 63 D5
/customer/riu_w 103e 64 D5
/customer/riu_w 103e 65 D5
/customer/riu_w 103e 66 D5
/customer/riu_w 103e 67 D5
/customer/riu_w 103e 68 D5
```

Q5: When using the dual-screen demoprog_panel program to test, the HDMI display is normal but the TTL or MIPI screen cannot be displayed?

The Panel display of prog_panel uses disp1, so it is necessary to configure the screen parameter m_wPanelDispPath of the screen used to 1, such as the SAT070AT50H18BH screen:

```
[SAT070AT50H18BH]
- m_wPanelDispPath = 0;
+ m_wPanelDispPath = 1;
m_pPanelName = "SAT070AT50H18BH_1024x600";
```

Q6: How to output the sensor to HDMI for display?

You can refer to the panel in the mi_demo directory, and use the screen parameters according to the README configuration in the directory.